

Dr. Dipankar Bhattacharya, Ph.D.

Marie Skłodowska-Curie Postdoctoral Fellow (2025–2027)

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Summary Statement

Knowledge as such does not have much power unless it is transformed into technology, which is capable of accelerating the progress of mankind. The long cherished dream of becoming the orchestrate of transforma- tion leads my eyes and heart towards the academics and research.

Education

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|-------------|---|
| 2016 – 2021 | ■ Ph.D., Mechatronics Enginnering in Department of Mechanical and Mechatronics Engineering, The University of Auckland (UoA), 01/08/2016 – 14/04/2021.
Thesis title: <i>Robotic Soft Esophagus Design, Modelling, and Control for Endoprosthetic Stent Testing and Food Swallow Investigation</i> |
| 2011 – 2013 | ■ M.Tech , Systems and Control in Department of Electrical Engineering, Indian Institute of Technology (IIT), 22/07/2011 – 04/07/2013.
Dissertation title: <i>Non-Linear Predictive Control Model using Relevance Vector Machine</i> . GPA: 8.4 / 10.0 |
| 2004 – 2010 | ■ B.Tech , Electronics and Communication Engineering , North Eastern Regional Institute of Science and Technology (NERIST), 21/07/2004 – 23/06/2010.
Project title: <i>Design and Synthesis of a Sixth Order Tow Thomas Bi-quad filter</i> . GPA: 9.1 / 10.0 |

Employment History (continued)

Employment History

Employment History (continued)

Supervising two Year 3 MEng (Design Engineering) placement students: **Ben Surtees** and **Lijiangke Liu** (March 2026 – Present). Res

Delivered lecture on **Robotics Simulation and Control using PyDrake** for *DESE71002 Robotics Research Projects*, Dysc

Employment History (continued)

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Teaching Experience

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Teaching Experience (continued)

Teaching Experience (continued)

Teaching Experience (continued)

Teaching Experience (continued)

Teaching Experience (continued)

BECE3304 *Neural Networks and Fuzzy Control*; **BECE2010** *Digital Electronics*; **BECE2016** *Signals and Systems*; **BTEE2002** *Network A*

Teaching Experience (continued)

Research Publications

Journal (Published/Accepted)

- 1 Jin*, B., Kobayashi*, A., Bhattacharya, D., Seino, A., Tokuda, F., Tien, N., & Kosuge, K. (2025). Automated Straight-line Sewing of Stretchable Fabrics with Different Lengths [*Equal contribution.]. *IEEE Robotics and Automation Letters*, 1–8. <https://doi.org/10.1109/LRA.2025.3619791>
- 2 Bhattacharya, D., Cheung, T. K., Wang, Y., & Lau, D. (2025). Kinematic and Dynamic Modeling of Cable-Object Interference and Wrapping in Complex Geometrical-Shaped Cable-Driven Parallel Robots. *Mechanism and Machine Theory*, 214, 106092. <https://doi.org/10.1016/j.mechmachtheory.2025.106092>
- 3 Bhattacharya, D., Chan, Y. P., Shang, S., Chan, Y. S., Tan, Y., & Lau, D. (2023). Tri-Space Operational Control of Redundant Multilink and Hybrid Cable-Driven Parallel Robots Using an Iterative-Learning based Reactive Approach. *IEEE Transactions on Control Systems and Technology*. <https://doi.org/10.1109/TCST.2023.3263689>
- 4 Bhattacharya, D., Cheng, L. K., & Xu, W. (2021). Nonlinear Model Predictive Control of a Robotic Soft Esophagus for Endoprosthetic Stent Testing. *IEEE Transactions on Industrial Electronics*. <https://doi.org/doi:10.1109/TIE.2021.3121755>

- 5 Bhattacharya, D., Jesna, S., Cheng, L. K., & Xu, W. (2021). RoSE: A Robotic Soft Esophagus for Endoprosthetic Stent Testing. *Soft Robotics*. <https://doi.org/10.1089/soro.2019.0205>
- 6 Dang, Y., Liu, Y., Hashem, R., Bhattacharya, D., Allen, J., Stommel, M., Cheng, L. K., & Xu, W. (2021). SoGut: A Soft Robotic Gastric Simulator. *Soft Robotics*. <https://doi.org/10.1089/soro.2019.0136>
- 7 Bhattacharya, D., Cheng, L. K., & Xu, W. (2020). Sparse Machine Learning Discovery of Dynamic Differential Equation of an Esophageal Swallowing Robot. *IEEE Transactions on Industrial Electronics*, 67(6), 4711–4720. <https://doi.org/10.1109/TIE.2019.2928239>
- 8 Bhattacharya, D., Nisha, M. G., & Pillai, G. (2015). Relevance vector-machine-based solar cell model. *International Journal of Sustainable Energy*, 34(10), 685–692. <https://doi.org/10.1080/14786451.2014.885030>
- 9 Dong, W., Bhattacharya, D., Kobayashi, A., Seino, A., Tokuda, F., Huang, X., Tang, K., Tien, N. C., & Kosuge, K. (2025). Precise Top-layer Fabric Segmentation for Fabric Destacking with Edge- and Shape-Aware Deep Networks. *2025 IEEE International Conference on Mechatronics and Automation (ICMA)*, 1343–1348.
- 10 Bhattacharya, D., Cheng, L. K., & Xu, W. (2018). Testing and Analysis of Migration Displacement of a Flared Stent Deployed in an Esophageal Swallowing Robot. *25th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*, 1–6. <https://doi.org/10.1109/M2VIP.2018.8600911>
- 11 Bhattacharya, D., Cheng, L. K., Dirven, S., & Xu, W. (2017). Actuation planning and modeling of a soft swallowing robot. *24th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*, 1–6. <https://doi.org/10.1109/M2VIP.2017.8211476>

Books and Chapters

- 1 Bhattacharya, D., Cheng, L. K., Dirven, S., & Xu, W. (2019). *Artificial intelligence approach to the trajectory generation and dynamics of a soft robotic swallowing simulator*. Springer International Publishing. https://doi.org/10.1007/978-3-319-78452-6_1

In Review/Preparation

- 1 Bhattacharya*, D., Tang*, K., Hang, X., Tokuda, F., Tien, N., & Kosuge, K. (2026). RTFF: Random-to-Target Fabric Flattening Policy Using Dual-Arm Manipulator, *IEEE International Conference on Robotics and Automation (ICRA)* [in review. *Equal contribution].
- 2 Wang, W., Bhattacharya, D., Kobayashi, A., Seino, A., Tokuda, F., Tien, N., & Kosuge, K. (n.d.). Automated Fabric Alignment Using Global-Local Optimization and Point Cloud Registration, *IEEE Transactions on Automation Science and Engineering (T-ASE)* [in review].
- 3 Cheung, T., Bhattacharya, D., & Lau, D. (n.d.). Rapidly Deployable Deformable Frame Cable-Driven Parallel Robots with Small Footprint [in preparation].
- 4 Wang, Y., Bhattacharya, D., & Lau, D. (n.d.). Static modelling and workspace analysis with arbitrary cable attachment design in a cable-driven parallel robot with passive backbone, *IEEE Transactions on Automation Science and Engineering* [in preparation].

Grants and Scholarships

Grants/Fellowships

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| 2025 |  Recipient, Horizon Europe Marie Skłodowska-Curie Postdoctoral Fellowship for the project "CASREx" (Grant No. 101205678), hosted at Imperial College London . Funded under HORIZON-MSCA-2024-PF-01-01. Duration: 24 months, EUR 276,187.92 . |
| 2023 |  Co-authored Innovation and Technology Fund (ITF), Partnerships Research Program (PRP) grant. |
| 2022 |  Co-authored Research Talent Hub for ITF project (RTH-ITF) grant. |

Grants and Scholarships (continued)

- 2018 ■ Recipient, Dynamic testing of implant stents using Robotic Soft Esophagus (RoSE), BvW Holding, Cham, Switzerland (39,000 NZD, 1 year) .
- 2018, 2019 ■ Recipient, Riddet Travel grant, Riddet Institute Centre of Research Excellence (3,000 NZD).

Scholarships

- 2019–2020 ■ Recipient, Doctor of Philosophy Scholarship, University of Auckland.
- 2016–2019 ■ Recipient, Doctor of Philosophy Scholarship, Riddet Institute Centre of Research Excellence.
- 2011–2013 ■ Recipient, Masters of technology Scholarship, Ministry of Human Resource (MHRD) GATE scholarship.

Academic Awards, and Miscellaneous Achievements

Academic Awards

- 15/12/25 ■ Selected, **European Talent Academy 2026** (Imperial College London, TU Munich, Politecnico di Milano).
- 2017 ■ Recipient, **Best Conference paper award**, 5th International Conference on Robotic Intelligence Technology and Applications (RiTA), Daejeon, Korea.
- **Ranked 2nd** in poster presentation in Delivery of Functionality in Complex Food Systems (DoF 2017) Conference.

Competitive Examinations

- 2011 ■ **Percentile: 99.12% (Rank 1,211/1,37,853)** in Graduate Aptitude Test of Engineering (GATE).
- 2010 ■ **Percentile: 96.00%** in Graduate Aptitude Test of Engineering (GATE).
- 2004 ■ **Ranked 31st** in All India Junior mathematics Olympiad.

Professional Activities

- 05/02/26 ■ Invited Speaker, **Dyson Engineering Departmental Seminar**, Imperial College London.
- 29/01/26 ■ Invited Speaker, **Imperial Robotics Postgrad Committee**, Imperial College London. Talk: *Learning-based Fabric Manipulation with Mesh Perception and Visual Servoing*.
- 28/01/26 ■ Delivered lecture on **Robotics Simulation and Control using PyDrake** for DESE71002 Robotics Research Projects, Dyson School of Engineering, Imperial College London.
- 08/08/25 ■ Invited Speaker, **World Robotics Conference (WRC 2025)**, Beijing, China. Talk: *Fabric modeling and perception for garment manufacturing*.
- 03/08/25 ■ Workshop Organizer and Presenter, **IEEE ICMA 2025**, Beijing, China. Talk: *Mesh-based Fabric State Estimation and Manipulation*.
- 29/05/23 ■ Poster Presentation, **IEEE ICRA 2023**, ExCeL London, United Kingdom.
- 2018 ■ Presenter, **BioMouth 2018 Meet**, Auckland, New Zealand.
- 2021 - Current ■ Reviewer of IEEE Robotics and Automation Letters (RAL), IEEE International Conference on Robotics and Automation (ICRA), ASME Journal of Mechanisms and Robotics (JMR).
- 2017 - Current ■ Member of the Institute of Electrical and Electronics Engineers (IEEE).
- 2014 ■ Former member of the Electrical Engineering Society (EES), IIT Roorkee, and NERIST Electronics Society (NES), NERIST.
- 2006 ■ Have an 'Outstanding Certificate' in National Service Scheme (NSS) for contributing in activities like blood donation camps, village cleanliness programmes and children education.

Skills

- Languages ■ English and Hindi: professional reading, writing, and speaking proficiency.

Skills (continued)

Research areas	■ Soft robotics, cable-driven robots, deformable-object manipulation, robot modelling and control, visual servoing, learning-based control, and robotic systems for healthcare and manufacturing.
Programming	■ Python, MATLAB, PyTorch, Gym framework, ROS, LabVIEW, assembly-level programming (8085, 8086, MASM32, and 8051), Git, and LaTeX.
Software	■ Blender, SOLIDWORKS, Autodesk Inventor, Inkscape, and Lightroom.
Hardware	■ Raspberry Pi, myRIO, and Arduino.

References

Prof Kazuhiro Kosuge

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